A Generic Simulation Framework for Warehouse Handling Systems with Automatic Model Generation

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Abstract
Warehouses decide where to put storage and pickup areas, how many docks and forklifts to run and which operating rules to follow, usually by rules of thumb. Each new layout normally needs a new simulation model, which is slow and requires an expert. We want to build one generic, parameter driven discrete event model of a warehouse, covering inbound receiving, putaway, order picking, staging and outbound shipping, that a non-expert can configure and run. On top of this framework we plan to develop an automatic model generation framework: the user draws a facility on a grid of standard building blocks and the complete simulation model is generated from it. The framework is used to size the forklift fleet, compare storage and pickup allotments and test policies under synthetic stochastic demand.

1. System Selection & Problem Statement

The System

A warehouse handling system: trucks arrive and wait in a yard, unload at a dock, pallets are checked and put away to storage, customer orders trigger picking, picked pallets are staged in pickup areas and loaded onto outbound trucks. The model is built in three components:

- **User Interface:** a non-expert sets layout, docks, forklifts, volume and policies, launches experiments and reads a KPI dashboard.
- **Generic Modelling Framework:** reusable building blocks (Dock, Forklift/charging, Passage), generic process logic and generic agents for trucks, pallets, orders and forklifts.
- **Data & Model Assembly:** stores configurations and input distributions, assembles the runnable model and exports results.

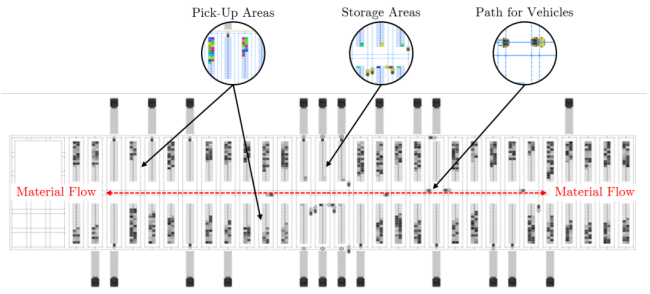


Figure 1. Simulation Model

Highlight: Automatic Model Generation

Building a new AnyLogic model for every layout is the main cost of simulation-based layout planning. We want to generate it automatically. The user fills a grid matrix where each cell is a building block (D = dock block, F = forklift block, P = passage, 0 = empty). A generator reads the matrix and creates the space markup, walls, docks, storage and pickup areas and the forklift path network at runtime. Any facility shape (I, L, T, U, H) then becomes a one-click experiment instead of a modelling task. This is built as an extension on top of the working framework.

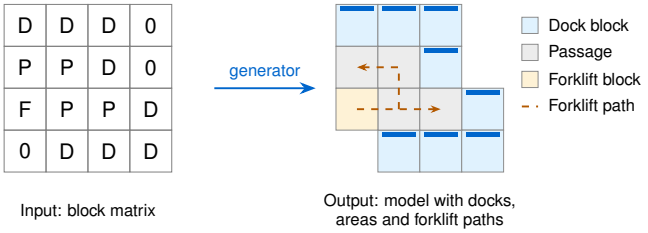


Figure 2. Automatic model generation: a block matrix is converted into a complete, runnable facility model.

The Problem

Poorly allotted storage and pickup areas and poorly routed material flow cause excess forklift travel, dock congestion and late truck departures. We study:

1. **Allotment & flow:** how should storage areas, pickup areas, vehicle paths and yard capacity be allotted to minimise travel and cycle time?
2. **Resources:** what minimum forklift fleet and dock count meets the service level at peak, and which resource is the bottleneck?
3. **Policies:** how much do dock, storage and dispatching rules improve performance at zero capital cost?
4. **Usability:** can one generic model, and ultimately one generated from a drawing, answer these without manual

rebuilding?

System Boundaries

Included	Excluded
Truck arrival, yard queue, docking	Gate check-in, yard manoeuvring
Unloading, receiving check, putaway	External road transport
Order arrival, picking, staging, loading	Replenishment, SKU slotting
Forklift travel, congestion, charging	Shifts and breaks
Dock, storage, dispatch policies	Returns, damages, IT

2. System Configuration

Entities & Resources

Entity	Attributes
Inbound truck	Arrival time, pallets, assigned dock
Outbound truck	Departure time, destination
Pallet	SKU class, storage location
Customer order	Lines per order, due time
Resource	Capacity / decision
Dock doors	Count; inbound share
Forklifts	Fleet size
Storage area	Slots; placement
Pickup / staging	Capacity; placement
Yard	Truck waiting slots
Chargers	Count

Process Flow Sketch

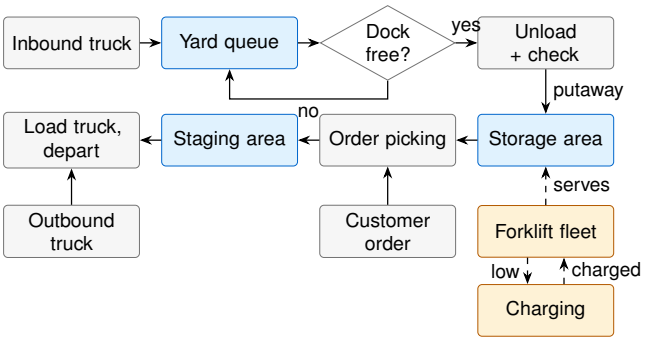


Figure 3. Routing logic. Blue: queues and areas with finite capacity; orange: forklift resource loop.

3. Input Data Collection Plan

All inputs use synthetic data. Distributions are fitted in Python, checked with KS / Chi-square tests, and the model is run on both sampled data and fitted distributions.

Stochastic variable	Sourcing	Sample / Dist.
Inbound truck inter-arrival	Synthetic	500; Exponential
Outbound departure time	Synthetic	200; Uniform
Pallets per truck	Synthetic	500; Empirical
Order inter-arrival	Synthetic	1000; Exponential
Lines per order	Synthetic	1000; Geometric
Unload / load time	Synthetic	500; Lognormal
Pick time per line	Synthetic	500; Lognormal
Forklift speed	Synthetic	Normal
Charge time	Synthetic	Uniform
Forklift MTBF / MTTR	Synthetic	200; Exp. / Lognormal
SKU class mix	Synthetic	Uniform vs. skewed

4. Output Measures & Optimization Goal

Key Performance Indicators

Category	KPI
Time	Cycle time, truck turnaround (mean, p95), yard wait
Throughput	Pallets, orders, trucks per shift
Utilisation	Forklifts, docks, storage, staging (%)
Distance	Total forklift travel distance
Congestion	Yard queue, forklift blocked time, staging overflow

The "What If" / Optimization Goal

Decision variable	Levels
Layout form	I, L, T; any shape via generation
Storage / pickup allotment	Near docks, centralised, SKU zones
Path direction	One-way, two-way
Docks (inbound share)	12, 18, 24 (25, 33, 50%)
Yard capacity	Low, medium, high
Forklift fleet size	Minimum feasible to saturation
Policies	Dock, storage, dispatch rules
Volume	Baseline, +20%, +40%

Optimization goal (OptQuest): minimise forklift fleet size subject to p95 truck turnaround $\leq T^*$ and zero staging overflow; then minimise forklift travel through allotment and policy choice.

References

[1] L. Jurgeleit, M. Mowe, M. Kiefer, C. Schumacher, and U. Clausen. Automatic model generation for discrete event simulation of less-than-truckload terminals. *Proc. Winter Simulation Conference*, 1493–1504, 2024.